

Global Path Planning Method based on Improved Beetle Antenna Search Algorithm for Unmanned Surface Vehicle

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Abstract: In this paper, the constraint function of the maximum turning angle is designed in the fitness function of the Beetle Antenna Search algorithm (BAS) according to the characteristic that unmanned surface vehicles (USV) are not easy to turn sharply. To solve the problem of strong randomness and poor stability of the global path planning of BAS, the dynamic search step size that changes with the distance from the obstacle is used in the Improved Beetle Antenna Search algorithm (IBAS), instead of the fixed step size used in the original algorithm. Simulation results show that IBAS improves the stability and success rate of global path planning for USV. The planned path is smooth, continuous, safe, and effective. IBAS is more suitable for the path planning of USV than the original algorithm.

Keywords: unmanned surface vehicle, Beetle Antenna Search algorithm, path planning, vessel collision avoidance

I. INTRODUCTION

In recent years, unmanned surface vehicles (USV) [1-2] have been incorporated into the battle sequence by many national navies, widely used to perform a variety of war and non-war military tasks, with independent planning, autonomous navigation capabilities, and advantages such as small size, high efficiency, intelligence, good concealment, wide activity area [3-5]. It plays an important role in environmental perception, target detection, mine sweeping and anti-submarine, amphibious operations, special operations, and port defense.

As one of the important algorithms for the navigation, guidance, and control system of USVs, global path planning can design routes that effectively avoid the blocking area and independently plan the optimal action path for safe navigation based on the task requirements and the known marine environment information [6]. It is one of the important research contents in the field of USVs [7] and an important embodiment of intelligent behavior [8].

In 2017, Jiang et al. proposed the Beetle Antenna Search algorithm (BAS), inspired by the behavior of longicorn beetles' foraging and courtship. The direction of movement is determined according to the odor concentration captured by the antennae on both sides of the longicorn beetle. The algorithm is simple in calculation and process, which can be optimized without gradient descent. It has obvious advantages over many new mainstream intelligent optimization algorithms [9-10].

In terms of path planning, the stability of BAS is poor for USVs, and USVs are not easy to turn sharply. The Improved

Beetle Antenna Search algorithm (IBAS) is proposed for the global path planning algorithm of USV, which changes the search step size with the environment to build a dynamic variable step size strategy to improve the stability of the algorithm. Moreover, the algorithm adds the constraint of the maximum turning angle, making the route more conducive to the navigation of USV.

II. IMPROVEMENT OF FITNESS FUNCTION

The original BAS has a large step size in the early stage of map search. If the size of obstacles encountered at this time is small, it is likely that the sampling point directly crosses the obstacles when searching the path. Although the sampling point does not hit the obstacles, the path formed by the connection between the current sampling point and the previous sampling point will pass through the obstacles, so this section of the path is invalid. By improving the fitness function of the original algorithm, invalid paths can be avoided. The fitness function expression for judging whether the path connected by the sampling points passes through the obstacle is shown in Equation 1.

$$\begin{aligned} \text{if } \{ \text{map}(X, Y) = 0 \mid (X, Y) \in L_{AB} \mid X \in (x_d, x), Y \in (y_d, y) \} \\ \text{fitness} = \sqrt{(x - x_g)^2 + (y - y_g)^2} \\ \text{else} \\ \text{fitness} = +\infty \end{aligned} \quad (1)$$

In Equation 1, $A(x_d, y_d)$ is the current position of the longicorn. $B(x, y)$ is the position of the antennae of the longicorn. $\text{map}(X, Y)$ is the map pixel matrix, in which 0 is the feasible area, and 1 is in the obstacle area. (x_g, y_g) is the position of the goal point.

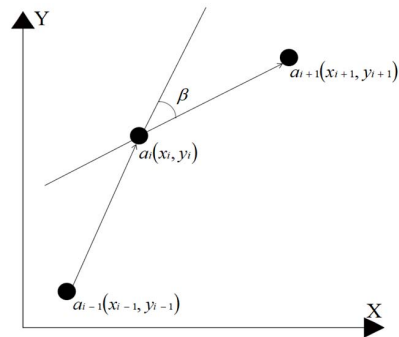


Figure 1. Turning angle of USVs.

Due to the large volume, large inertia, and the interference of wind, wave, current, and other external factors on the water, the USV can not complete a large turning in a short time. Therefore, it is necessary to limit the turning angle of the USV in the process of path search. The maximum turning angle of the USV is shown in Equation 2.

$$\beta = \arccos \left(\frac{(x_i - x_{i-1})(x_{i+1} - x_i) + (y_i - y_{i-1})(y_{i+1} - y_i)}{\sqrt{(x_i - x_{i-1})^2 + (y_i - y_{i-1})^2} \sqrt{(x_{i+1} - x_i)^2 + (y_{i+1} - y_i)^2}} \right) \quad (2)$$

In Equation 2, (x_i, y_i) is the current position of the longicorn, as shown in Figure 1. (x_{i-1}, y_{i-1}) is the position of the previous step of the longicorn. (x_{i+1}, y_{i+1}) is the position of the next step for searching.

In the search for planning path, if the turning angle of the longicorn beetle is greater than the limited maximum turning angle $\beta_{\max}$ when it reaches the next point, a penalty function is added to the fitness function to avoid excessive turning angle of path search, as shown in Equation 3.

$$f(x) = \begin{cases} \sqrt{(x - x_g)^2 + (y - y_g)^2} & \text{abs}(\beta) \leq \beta_{\max} \\ +\infty & \text{abs}(\beta) > \beta_{\max} \end{cases} \quad (3)$$

Based on the improved fitness function, when the algorithm calculates the value of the fitness function of longicorn beetles, not only the current position and the next longicorn beetles' position are considered, but also the historical position of the previous step is also considered to calculate the turning angle. If the turning angle exceeds the set maximum turning angle, the value of the fitness function becomes infinite. Therefore, the longicorn in this position is excluded from the search decision. When the longicorn beetles search in the global map, filter the longicorn beetles according to the fitness function. First, exclude the longicorn beetles with the turning angle more than $\beta_{\max}$, and then filter out the longicorn beetles with the best fitness value from the remaining longicorn beetle population whose turning angle is less than $\beta_{\max}$ for the next step length search.

III. IMPROVEMENT OF STEP SIZE

The original BAS uses a fixed step size or a dynamic attenuation step size based on the distance to the target point for searching, but this is not applicable to the path planning problem. In the path planning problem, there is the possibility of obstacles at any position between the start point and the target point. Moreover, the adoption of the global dynamic attenuation step without considering obstacles will increase the possibility that the longicorn beetle will fall into local extreme value. If the longicorn beetle falls into the local extreme value, large steps should be taken to jump out of the local extreme value area. Therefore, the improvement of the dynamic step size is proposed in the IBAS, which is based on the distance to the obstacle. The improvement plan is as follows:

(1) When the distance between the longicorn beetle and the obstacle is more than twice the initial step size, it is judged that the longicorn beetle is in an open area, and the longicorn beetle conducts path exploration with the initial step size.

(2) When the distance between the longicorn beetle and the obstacle is less than twice the initial step size, it is judged that the longicorn beetle is in the area affected by the obstacle, and the step size decreases with the decrease of the distance from the obstacle, thus improving the local search accuracy.

(3) When the distance between the longicorn beetle and the obstacle is less than 0.5 times the initial step size, if the step size continues to decay at this time, the probability of the longicorn beetle falling into the local extreme value will be greatly increased. At this time, the step size mutation will be conducted with twice the initial step size, and large step size exploration will be used to reduce the probability of the longicorn beetle falling into the local extreme value area.

$$\text{step} = \begin{cases} \text{step0} & d_{\text{obs}} \geq 2 \times \text{step0} \\ \text{step0} \times \text{eta} & 0.5 \times \text{step0} \leq d_{\text{obs}} < 2 \times \text{step0} \\ 2 \times \text{step0} & d_{\text{obs}} < 0.5 \times \text{step0} \end{cases} \quad (4)$$

In Equation 4, step0 is the initial step size. d_{obs} is the distance between the longicorn beetle and the obstacle. eta is the dynamic attenuation coefficient of step size in the original BAS.

Through the above three steps, the original fixed step size is changed into a dynamic variable step size based on the distance between the obstacles. The initial step size becomes the only variable in the dynamic step size strategy, so the selection of the initial step size greatly affects the success of path planning. When selecting the initial step size, the step size should not exceed the distance between the starting point and the target point. Otherwise, the optimal solution will not be found. In addition, the limit of the turning angle will also affect the step size. Too large step size in the multi-extremum region may cause the turning angle of the longicorn to exceed the constraint range, so it is necessary to limit the maximum step size in the search process. Therefore, in IBAS, the selection range of initial step size includes $(0, 0.05d_{\text{goal}}]$, $(0.05d_{\text{goal}}, 0.1d_{\text{goal}}]$, $(0.1d_{\text{goal}}, 0.15d_{\text{goal}}]$, and $(0.15d_{\text{goal}}, 0.2d_{\text{goal}}]$. The range of initial step size should be selected according to the map.

IV. SIMULATION

There are the simulations of path planning for the USV using IBAS we proposed, compared with the original BAS [11]. The map is 600 pixels $\times$ 600 pixels, and the number of the longicorn is 40. The simulation diagram of BAS and IBAS are shown in Figure 2.~Figure 7.

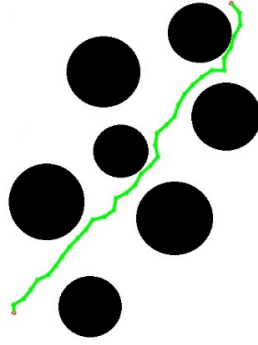


Figure 2. The planned path of BAS.

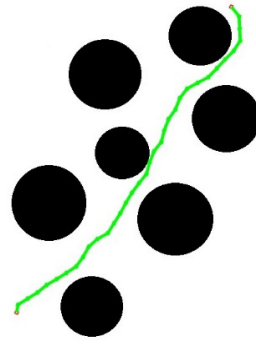


Figure 3. The planned path of IBAS.

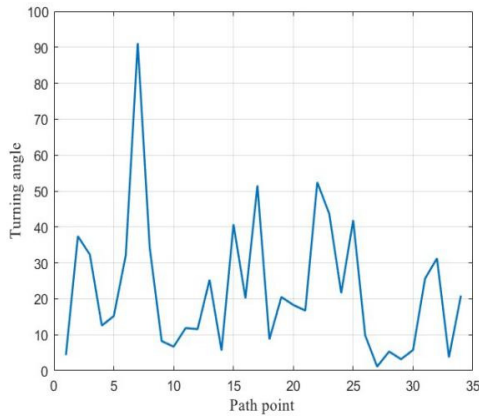


Figure 4. The turning angle of BAS.

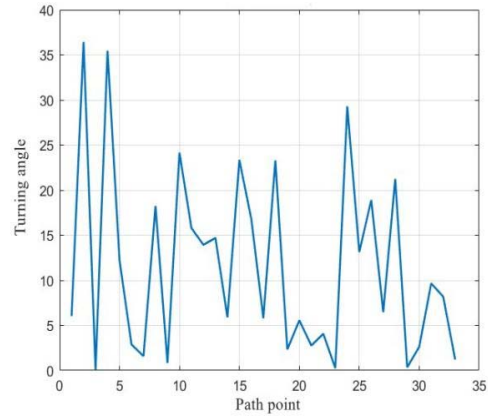


Figure 5. The turning angle of IBAS.

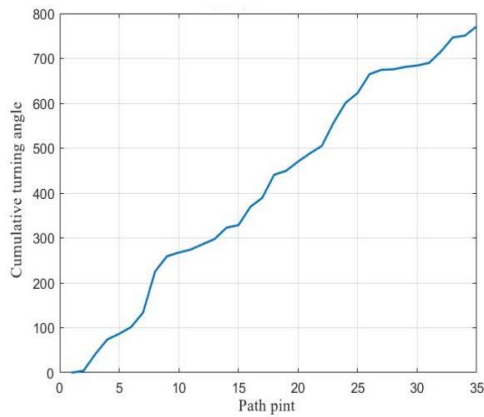


Figure 6. The cumulative turning angle of BAS.

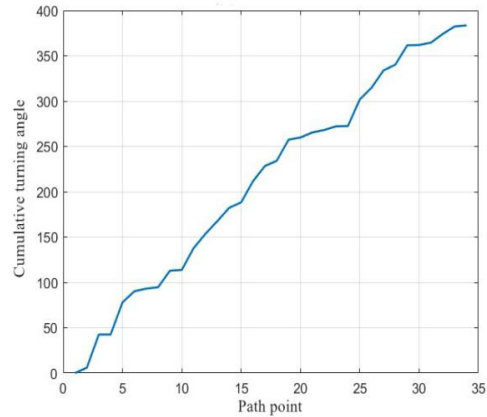


Figure 7. The cumulative turning angle of IBAS.

The simulation results of the original BAS and IBAS are shown in Table 1.

Table 1. The simulation results of BAS and IBAS

Algorithm	Path length	Max Turning angle	Cumulative turning angle
BAS	656.7742	92°	774°
IBAS	639.4769	36°	381°

In the simulation results, the max turning angles of BAS and IBAS are 92° and 36° respectively. The cumulative turning angles of BAS and IBAS are 774° and 381° respectively, as shown in Table 1. It is difficult for USVs sailing on the water to complete more than 90° turning in a short time, and the cumulative turning angle and the path length of IBAS are less than these of BAS. Therefore, IBAS is more suitable for the path planning of the USV.

In this simulation, when the initial step size is equal to 0.05

times the distance between the start point and the target point, 11 times of path planning are successful, but all of them fall into the local extremum and jump out of the local extremum, which reduces the efficiency of path planning. When using 0.1 times the distance between the start point and the target point, all 20 times of path planning were successful, and one of them fell into the local extreme and jumped out. When using 0.15 times the distance between the start point and the target point, the 20 times of path planning were all successful, and none of them fell into the local extreme value. When using 0.2 times the distance between the start point and the target point, the 20 times of path planning failed. Therefore, the range of initial step size selection is between 0.1 and 0.15 times the distance between the start point and the target point to improve the success probability of path planning of IBAS.

V. CONCLUSION

In this paper, the maximum turning angle constraint is added to IBAS based on the motion characteristics of the USV, which is not easy to turn sharply. The dynamic search step size is improved in IBAS, based on the distance between the longicorn and the obstacle. The step size is attenuating when the longicorn approaches the obstacle. When the distance between the longicorn and the obstacle is reduced to a certain value, the step size is mutated to prevent it from falling into the local extreme. Through the simulation and analysis, the cumulative turning angle and the path length of IBAS are less than these of the original BAS. IBAS has more success probability of path planning, when the range of initial step size selection is between 0.1 and 0.15 times the distance between the start point and the target point. Therefore, IBAS is more suitable for the path planning of the USV.

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